

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's output ontology is a four-class A/B/C/D classification [Q83, Q44] with classification accuracy as the sole performance criterion [Q34, Q23]. Dataset analysis confirms a single ClassLabel field (0–3) with no rationale field. The deployment requires a fundamentally richer output taxonomy: a verdict (sahi/galat) plus a substantive Hindi-language explanation in standard Manak Hindi [elicitation Q3]. MMLU's design provides zero signal on the model's capacity to produce explanatory rationale, let alone in Hindi. The benchmark's stated rationale for the simple format ('a simple-to-evaluate test that measures classification accuracy' [Q23]) is internally coherent but structurally incompatible with the deployment's pedagogical-feedback requirement.

ID	Evidence
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q23	"Consequently, we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q3	"We find that while most recent models have near random-chance accuracy, the very largest GPT-3 model improves over random chance by almost 20 percentage points on average." (p.1)

Strengths

- Verdict component (correct vs. incorrect) is a partial match — the A/B/C/D classification can be reduced to a binary correct/incorrect signal that does align with one component of the deployment output (the sahi/galat label) [Q83, DATASET concern 3].

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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values

OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels (A/B/C/D) [Q83] partially match the deployment's required correctness verdict but entirely omit the explanation component. Deployment requires Hindi explanatory feedback per elicitation Q3.

OO-2: Missing categories: free-form Hindi explanation, distractor analysis (why each wrong option is wrong), and reference to canonical Indian sources (NCERT, Laxmikant). None are represented in MMLU's output schema.

OO-3: Output categories themselves are content-neutral (letter labels), but the underlying scoring rule treats US-anchored answer keys (e.g., moral_scenarios under US 2020 norms [DATASET-D18, D19]) as correctness ground truth, encoding non-regional values.

OO-4: A stakeholder-driven output taxonomy redesign is required: minimum {verdict_label, hindi_explanation_text, optional_distractor_analysis} with a separate rubric for explanation quality. No existing benchmark covers this — the rubric gap is documented as unresolved [WEB-21].

OO-5: Structural and content validity are severely harmed: MMLU's output structure measures only one of two required output components and applies decision rules anchored in US norms.

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Q3	"We find that while most recent models have near random-chance accuracy, the very largest GPT-3 model improves over random chance by almost 20 percentage points on average." (p.1)
Q23	"Consequently, we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
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D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)
Q62	"Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining." (p.8)

Information Gaps

- No published rubric for Hindi exam-prep explanation quality [WEB-21] — cannot quantify the magnitude of the output-ontology gap.

Requires Expert Verification

- Indian exam-prep pedagogues need to define an explanation-quality rubric (factual accuracy, distractor analysis, Manak Hindi register, NCERT alignment) for the deployment.

Remediation

Gap 1: Label-only A/B/C/D output cannot evaluate the required Hindi explanatory feedback.

Recommendation: Commission a Hindi exam-prep explanation-quality rubric (factual accuracy, distractor analysis, Manak Hindi register, NCERT alignment) with Indian SME panel — this is a documented gap in current literature [WEB-21] and must be built bespoke for the deployment.

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